

the chain rule of probability

CSCI
270

can we express this
more compactly?

joint distribution	$X_{\text{♀}}$	$X_{\text{♂}}$	X	$Y_{\text{♀}}$	$Y_{\text{♂}}$	Y	$Z_{\text{♀}}$	$Z_{\text{♂}}$	Z	
			A			A			A	p_1
			A			A			B	p_2
	⋮									⋮
			0			0			0	p_{46656}

46656 rows

idea:

re-express
joint probability
as a product of
conditional
probabilities

$$P(x_2 | x_1) \overset{y|x}{=} \frac{P(x_1, x_2)}{P(x_1)}$$

so: $P(x_1, x_2) = P(x_2 | x_1) P(x_1)$

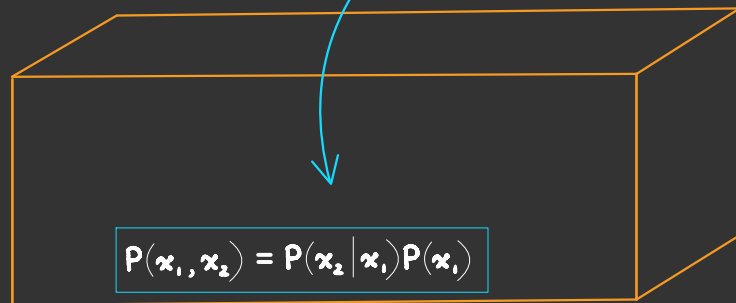
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box of results

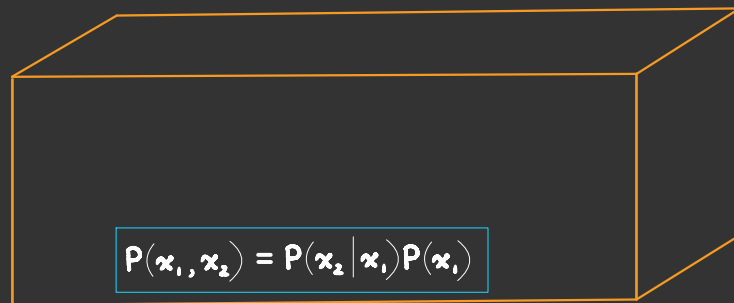
idea:

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$$P(x_3 | x_1, x_2) \stackrel{y|x}{=} \frac{P(x_1, x_2, x_3)}{P(x_1, x_2)}$$

so:

$$P(x_1, x_2, x_3) = P(x_3 | x_1, x_2) P(x_1, x_2)$$



box of results

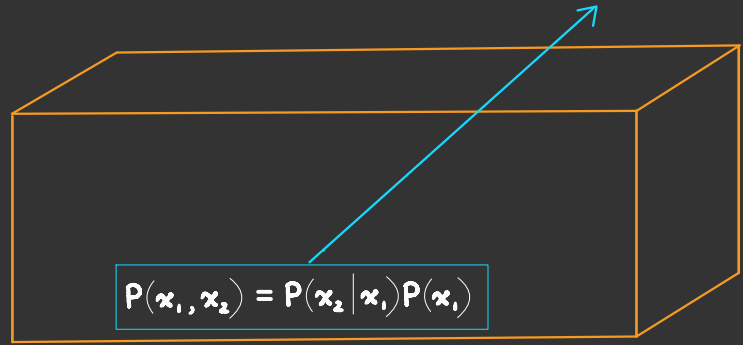
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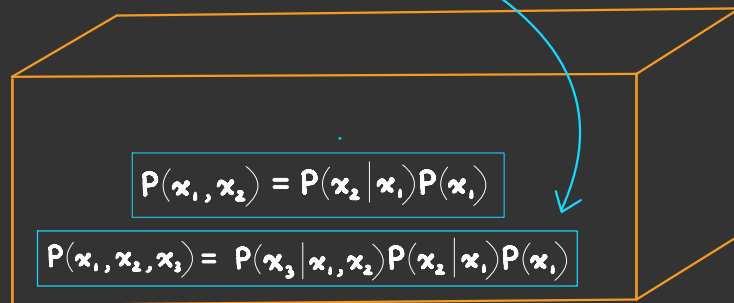
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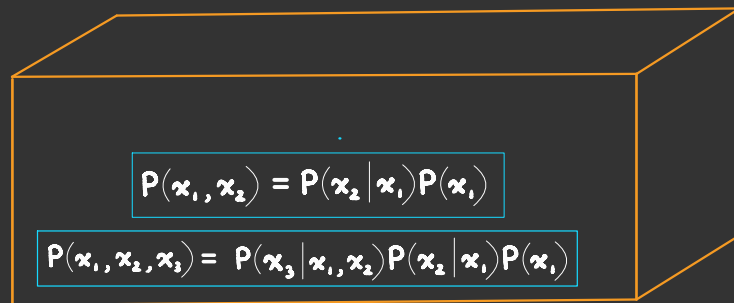
box of results

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$$P(x_4 | x_1, x_2, x_3) \stackrel{y|x}{=} \frac{P(x_1, x_2, x_3, x_4)}{P(x_1, x_2, x_3)}$$

so:

$$P(x_1, x_2, x_3, x_4) = P(x_4 | x_1, x_2, x_3) P(x_1, x_2, x_3)$$



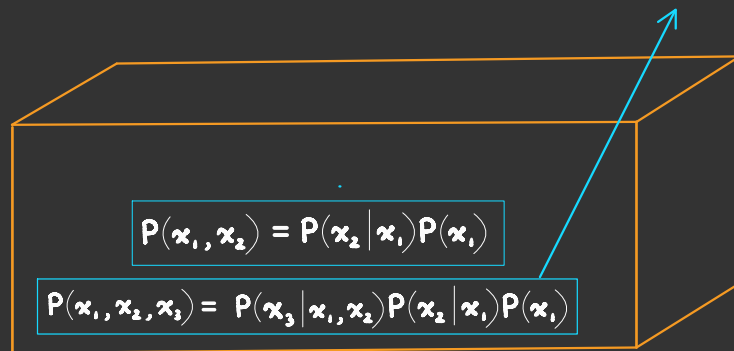
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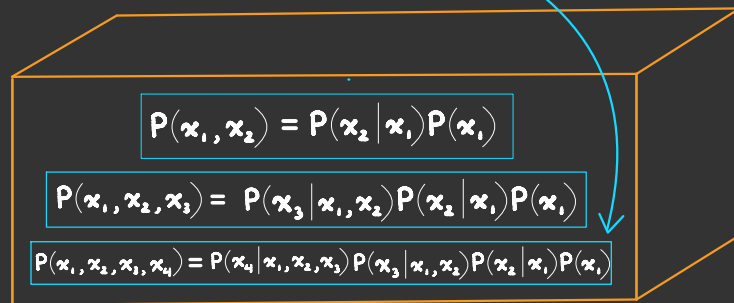
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box of results

what is the general rule?

$$P(x_1, x_2) = P(x_2 | x_1) P(x_1)$$

$$P(x_1, x_2, x_3) = P(x_3 | x_1, x_2) P(x_2 | x_1) P(x_1)$$

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⋮

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | x_1, \dots, x_{i-1})$$



the chain rule
of probability

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid x_1, \dots, x_{i-1})$$